

AZ-104 Microsoft Azure Administrator Exam Study Guide

Passing this exam earns the Microsoft Certified: Azure Administrator Associate certification. The AZ-104 tests your ability to implement, manage, and monitor Azure environments across identity, governance, storage, compute, and networking.

[Microsoft Learn](#) This comprehensive guide covers all five exam domains with their official weightages, including CLI commands, PowerShell cmdlets, critical limits, and comparison tables essential for exam success.

Exam overview and structure

Domain	Weightage	Focus Areas
Manage Azure identities and governance	20-25%	Entra ID, RBAC, Policy, Management Groups
Implement and manage storage	15-20%	Storage accounts, SAS, redundancy, tiers
Deploy and manage Azure compute resources	20-25%	VMs, VMSS, containers, App Service, ARM/Bicep
Implement and manage virtual networking	15-20%	VNets, NSGs, Load Balancers, DNS, Bastion
Monitor and maintain Azure resources	10-15%	Azure Monitor, alerts, backup, Site Recovery

Exam Details: 40-60 questions, 120 minutes, **700 points** minimum to pass (scale 100-1000) [Pluralsight](#)

Domain 1: Manage Azure identities and governance (20-25%)

Microsoft Entra ID users and groups

Core terminology

Term	Definition
Tenant	Dedicated instance of Microsoft Entra ID representing an organization
User Principal Name (UPN)	Unique identifier format: <u>user@domain.com</u>
Security Group	Used for managing access to resources; contains users, devices, service principals
Dynamic Group	Membership automatically managed based on user/device attributes
Guest User (B2B)	External user invited using their own credentials

CLI commands for user management

bash

Create a new user

```
az ad user create --display-name "John Smith" \
  --user-principal-name "jsmith@contoso.com" \
  --password "SecureP@ssw0rd!" \
  --force-change-password-next-sign-in true
```

Create a security group

```
az ad group create --display-name "Marketing Team" --mail-nickname "MarketingTeam"
```

Add member to group

```
az ad group member add --group "Marketing Team" --member-id <user-object-id>
```

List all users

```
az ad user list --output table
```

PowerShell cmdlets for user management

```
powershell
```

```
# Create new user
```

```
$passwordProfile = New-Object -TypeName Microsoft.Open.AzureAD.Model.PasswordProfile
```

```
$passwordProfile.Password = '<Strong-Password>'
```

```
New-EntraUser -DisplayName 'New User' -PasswordProfile $passwordProfile `
  -UserPrincipalName 'NewUser@contoso.com' -AccountEnabled $true -MailNickName 'NewUser'
```

```
# Create security group
```

```
New-EntraGroup -DisplayName 'Contoso Marketing' -MailEnabled $false `
```

```
-SecurityEnabled $true -MailNickName 'ContosoMarketing'
```

```
# Add user to group
```

```
Add-EntraGroupMember -GroupId $group.Id -MemberId $user.Id
```

Self-Service Password Reset (SSPR)

Authentication methods available: Mobile app notification/code, Email, Mobile phone (SMS), Office phone, Security questions (not available for admins)

Feature	License Required
Cloud-only password change	Free
Cloud-only password reset	Microsoft Entra ID P1/P2
Password writeback to on-premises	Microsoft Entra ID P1/P2

Key point: Administrators always have SSPR enabled with **two-gate policy** requiring two authentication methods.

Azure RBAC and access management

Built-in role hierarchy

Role	Permissions	Role ID
Owner	Full access + can assign roles	8e3af657-a8ff-443c-a75c-2fe8c4bcb635
Contributor	Full access, cannot assign roles	b24988ac-6180-42a0-ab88-20f7382dd24c
Reader	View all resources only	acdd72a7-3385-48ef-bd42-f606fba81ae7
User Access Administrator	Manage user access to resources	18d7d88d-d35e-4fb5-a5c3-7773c20a72d9

Role assignment commands

bash

Assign role at subscription scope

```
az role assignment create --assignee "user@contoso.com" --role "Reader" \  
  --subscription <subscription-id>
```

Assign role at resource group scope

```
az role assignment create --assignee "user@contoso.com" --role "Contributor" \  
  --resource-group myResourceGroup
```

List role assignments for a user

```
az role assignment list --assignee user@contoso.com --all --output table
```

powershell

Assign role at resource group scope

```
New-AzRoleAssignment -SignInName "user@contoso.com" -RoleDefinitionName "Contributor" \  
  -ResourceGroupName "myResourceGroup"
```

Get all role assignments including inherited from groups

```
Get-AzRoleAssignment -SignInName "user@contoso.com" -ExpandPrincipalGroups
```

RBAC scope hierarchy

Management Group → Subscription → Resource Group → Resource

Critical concept: Permissions **inherit downward**—roles assigned at parent scope apply to all child resources. (Microsoft Learn) Effective permissions = sum of all role assignments (additive model).

Azure Policy and governance

Policy effects

Effect	Behavior
Deny	Block non-compliant resource creation/modification
Audit	Log non-compliant resources without blocking
Append	Add fields to resources during creation
Modify	Add, update, or remove tags/properties
DeployIfNotExists	Deploy resources if condition met
AuditIfNotExists	Audit if related resource doesn't exist
Disabled	Policy not evaluated

Policy CLI commands

```
bash
```

```
# Create policy assignment with parameters
```

```
az policy assignment create --name 'allowed-locations' \  
  --scope '/subscriptions/{sub-id}' \  
  --policy '/providers/Microsoft.Authorization/policyDefinitions/e56962a6-4747-49cd-b67b-bf8b0197' \  
  --params '{"listOfAllowedLocations":{"value":["eastus","westus"]}}'
```

```
# Get compliance state
```

```
az policy state list --resource-group myResourceGroup --output table
```

Resource locks

Lock Type	Effect
CanNotDelete	Resources can be read and modified but not deleted
ReadOnly	Resources can only be read; no modifications or deletions

```
bash
```

```
# Create delete lock on resource group
```

```
az lock create --name "PreventDelete" --lock-type CanNotDelete --resource-group myResourceGroup
```

```
# Create read-only lock
```

```
az lock create --name "ReadOnlyLock" --lock-type ReadOnly --resource-group myResourceGroup
```

Important: Locks apply to **management plane operations only**, not data plane. A ReadOnly lock on a storage account blocks the **List Keys** operation. [Microsoft Learn](#)

Tags

Limits: Maximum **50 tags** per resource/resource group/subscription. [Microsoft Learn](#) Tag name: 512 chars max. Tag value: 256 chars max. Tags do **NOT inherit** from resource groups to resources. [Microsoft Learn](#)

```
bash
```

```
# Merge tags (add to existing)
```

```
az tag update --resource-id <resource-id> --operation Merge --tags CostCenter=00123
```

```
# Replace all tags
```

```
az tag update --resource-id <resource-id> --operation Replace --tags NewTag=NewValue
```

Management groups

Hierarchy depth: Up to **6 levels** (excluding root and subscription level)

```
bash
```

```
# Create management group
```

```
az account management-group create --name "Contoso" --display-name "Contoso Group"
```

```
# Move subscription to management group
```

```
az account management-group subscription add --name "Contoso" --subscription {subscription-id}
```

Domain 1 limits reference

Resource	Limit
Role assignments per subscription	4,000
Custom roles per tenant	5,000
Management groups per tenant	10,000
Policy definitions per subscription	500
Policy assignments per scope	200
Tags per resource	50

Domain 2: Implement and manage storage (15-20%)

Storage account configuration

Storage account types

Type	Use Case	Supported Services
Standard GPv2	Recommended for most scenarios	Blob, File, Queue, Table
Premium Block Blobs	High transaction rates, low latency	Block blobs, append blobs
Premium File Shares	High-performance file shares	Azure Files (SSD)
Premium Page Blobs	Unmanaged VM disks	Page blobs

Create storage account

bash

Create storage account with security settings

```
az storage account create --name <storage-account> --resource-group <rg> \  
  --location <location> --sku Standard_LRS --kind StorageV2 \  
  --min-tls-version TLS1_2 --allow-blob-public-access false
```

powershell

```
New-AzStorageAccount -ResourceGroupName <rg> -Name <account-name> -Location <location> \  
  -SkuName Standard_RAGRS -Kind StorageV2 -AllowBlobPublicAccess $false -MinimumTlsVersion
```

Storage redundancy options

Option	Copies	Primary Region	Secondary Region	Durability	Read Access Secondary
LRS	3	Single zone	None	11 nines	No
ZRS	3	3 availability zones	None	12 nines	No
GRS	6	LRS (3)	LRS (3)	16 nines	No
RA-GRS	6	LRS (3)	LRS (3)	16 nines	Yes
GZRS	6	ZRS (3 zones)	LRS (3)	16 nines	No
RA-GZRS	6	ZRS (3 zones)	LRS (3)	16 nines	Yes

Critical points:

- Archive tier only supported for **LRS, GRS, RA-GRS** (NOT ZRS, GZRS, RA-GZRS)
- Azure Files doesn't support RA-GRS or RA-GZRS ([Microsoft Learn](#))
- Premium file shares support only LRS and ZRS
- Secondary endpoint: `<account>-secondary.blob.core.windows.net`

Access tiers comparison

Tier	Min Retention	Storage Cost	Access Cost	Latency	Use Case
Hot	None	Highest	Lowest	Milliseconds	Frequently accessed data
Cool	30 days	Lower	Higher	Milliseconds	Infrequently accessed
Cold	90 days	Lower	Higher	Milliseconds	Rarely accessed
Archive	180 days	Lowest	Highest	Hours (rehydration)	Long-term backup, compliance

Archive tier: Metadata readable, blob content offline. Rehydration takes up to **15 hours** (standard priority).

```
bash

# Change blob tier
az storage blob set-tier --account-name <account> --container-name <container> \
  --name <blob> --tier Archive

# Set default account tier
az storage account update --name <account> --resource-group <rg> --access-tier Cool
```

Shared Access Signatures (SAS)

SAS types comparison

SAS Type	Description	Signed With	Scope
User Delegation SAS	Most secure, Blob storage only	User delegation key	Blob containers/ blobs
Service SAS	Delegates access to one service	Storage account key	Single service
Account SAS	Delegates access to multiple services	Storage account key	Multiple services

```
bash
```

Generate container SAS

```
az storage container generate-sas --account-name <account> --name <container> \  
--permissions rwdl --expiry <datetime>
```

Generate account SAS

```
az storage account generate-sas --account-name <account> --account-key <key> \  
--expiry <datetime> --permissions rwdlacup --resource-types sco --services bfqt
```

Stored Access Policies: Up to **5 per container/share/queue/table**. Deleting a policy revokes all SAS tokens referencing it.

AzCopy essential commands

```
bash
```

Login with Microsoft Entra ID

```
azcopy login
```

Upload file

```
azcopy copy '<local-file>' 'https://<account>.blob.core.windows.net/<container>/<blob>'
```

Upload directory recursively

```
azcopy copy '<local-directory>' 'https://<account>.blob.core.windows.net/<container>' --recursive
```

Upload to specific tier

```
azcopy copy '<local-file>' 'https://<account>.blob.core.windows.net/<container>/<blob>' --block-l
```

Copy between storage accounts

```
azcopy copy 'https://<source>.blob.core.windows.net/<container>/<blob>' \  
'https://<dest>.blob.core.windows.net/<container>/<blob>'
```

Sync local to blob (one-way)

```
azcopy sync '<local-directory>' 'https://<account>.blob.core.windows.net/<container>' --recursive
```

Soft delete and versioning

```
bash
```

```
# Enable blob soft delete (1-365 days)
```

```
az storage blob service-properties delete-policy update --account-name <account> \  
--enable true --days-retained 14
```

```
# Enable container soft delete
```

```
az storage account blob-service-properties update --account-name <account> \  
--resource-group <rg> --container-delete-retention-days 7 --enable-container-delete-retention true
```

```
# Enable blob versioning
```

```
az storage account blob-service-properties update --account-name <account> \  
--resource-group <rg> --enable-versioning true
```

Domain 2 limits reference

Resource	Limit
Storage accounts per region per subscription	250 (500 with quota increase) Microsoft Learn
Maximum storage account capacity	5 PiB
Maximum blob size (block blob)	~190.7 TiB
Standard file share size	5 TiB default, 100 TiB with large files
Premium file share size	100 TiB
Maximum file size	4 TiB
Stored access policies per container	5

Domain 3: Deploy and manage Azure compute resources (20-25%)

ARM templates and Bicep

ARM template structure

json

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json",
  "contentVersion": "1.0.0.0",
  "parameters": { },
  "variables": { },
  "functions": [ ],
  "resources": [ ],
  "outputs": { }
}
```

Element	Required	Max Count
\$schema	Yes	—
contentVersion	Yes	—
parameters	No	256
variables	No	256
resources	Yes	800
outputs	No	64

Bicep syntax example

bicep

```
// Parameters
param location string = resourceGroup().location
param storageAccountName string

// Variables
var uniqueStorageName = '${storageAccountName}${uniqueString(resourceGroup().id)}'

// Resources
resource storageAccount 'Microsoft.Storage/storageAccounts@2023-01-01' = {
  name: uniqueStorageName
  location: location
  sku: { name: 'Standard_LRS' }
  kind: 'StorageV2'
}

// Outputs
output storageEndpoint string = storageAccount.properties.primaryEndpoints.blob
```

Deployment commands

```
bash
```

```
# Deploy to resource group
```

```
az deployment group create --resource-group <rg> --template-file <template.json|bicep>
```

```
# What-if preview
```

```
az deployment group what-if --resource-group <rg> --template-file <path>
```

```
# Convert ARM to Bicep
```

```
az bicep decompile --file template.json
```

```
# Build Bicep to JSON
```

```
az bicep build --file main.bicep
```

Virtual machines

VM size families

Family	Series	Use Case
General Purpose	B-series	Burstable, variable workloads
General Purpose	D-series	Enterprise apps, production
Memory Optimized	E-series	Databases, high memory-to-CPU
Compute Optimized	F-series	Batch processing, high CPU
Storage Optimized	L-series	Big Data, high disk throughput
GPU	N-series	ML, rendering

Create and manage VMs

```
bash
```

```
# Create Linux VM with SSH keys
```

```
az vm create --resource-group myResourceGroup --name myVM --image Ubuntu2204 \  
  --admin-username azureuser --generate-ssh-keys --size Standard_D2s_v5
```

```
# Create VM in availability zone
```

```
az vm create --resource-group myResourceGroup --name myVM --image Ubuntu2204 --zone 1
```

```
# Resize VM
```

```
az vm resize --resource-group myResourceGroup --name myVM --size Standard_D4s_v5
```

```
# Add data disk
```

```
az vm disk attach --resource-group myResourceGroup --vm-name myVM --name myDataDisk \  
  --size-gb 128 --sku Premium_LRS --new
```

Availability options comparison

Feature	Availability Set	Availability Zone	VM Scale Set
SLA	99.95%	99.99%	Varies
Scope	Single datacenter	Multiple datacenters	Region/Zones
Fault Domains	Up to 3	1 per zone	Configurable
Update Domains	Up to 20	1 per zone	Automatic
Auto-scaling	No	No	Yes

```
bash
```

```
# Create availability set
```

```
az vm availability-set create --resource-group myResourceGroup --name myAvailabilitySet \  
  --platform-fault-domain-count 3 --platform-update-domain-count 5
```

```
# Create VM Scale Set
```

```
az vmss create --resource-group myResourceGroup --name myScaleSet --image Ubuntu2204 \  
  --orchestration-mode Flexible --instance-count 2 --vm-sku Standard_D2s_v5 --zones 1 2 3
```

Containers

Azure Container Registry SKUs

SKU	Storage	Webhooks	Geo-replication	Private Link
Basic	10 GB	2	No	No
Standard	100 GB	10	No	Yes
Premium	500 GB	500	Yes	Yes

```
bash
```

```
# Create ACR
```

```
az acr create --resource-group myResourceGroup --name myRegistry --sku Standard
```

```
# Build and push image
```

```
az acr build --registry myRegistry --image myapp:v1 .
```

Azure Container Instances

```
bash
```

```
# Create container instance
```

```
az container create --resource-group myResourceGroup --name mycontainer \
  --image mcr.microsoft.com/azuredocs/aci-helloworld --ports 80 \
  --dns-name-label myapp --ip-address public --cpu 2 --memory 4
```

ACI Limits: Container image: 15 GB max, Memory: 16 GB per group, CPU: 4 cores per group

Azure Container Apps

```
bash
```

```
# Create Container Apps environment
```

```
az containerapp env create --name myEnvironment --resource-group myResourceGroup --location e
```

```
# Create container app with scaling
```

```
az containerapp create --name myapp --resource-group myResourceGroup --environment myEnviro
  --image mcr.microsoft.com/azuredocs/containerapps-helloworld:latest \
  --target-port 80 --ingress external --min-replicas 0 --max-replicas 10
```

Azure App Service

App Service Plan tiers

Tier	Features	Max Instances	Deployment Slots
Free (F1)	Shared compute, 60 min/day	N/A	0
Basic (B1-B3)	Dedicated VMs, manual scaling	3	0
Standard (S1-S3)	Autoscale, staging slots	10	5
Premium (P1v3-P3v3)	Enhanced performance	30	20
Isolated (I1-I3)	Private environment (ASE)	100	20

App Service commands

```
bash

# Create App Service Plan
az appservice plan create --name myAppServicePlan --resource-group myResourceGroup \
  --sku S1 --is-linux

# Create Web App
az webapp create --name myWebApp --resource-group myResourceGroup \
  --plan myAppServicePlan --runtime "DOTNET|6.0"

# Create deployment slot
az webapp deployment slot create --resource-group myResourceGroup --name myWebApp --slot sta

# Swap slots
az webapp deployment slot swap --resource-group myResourceGroup --name myWebApp \
  --slot staging --target-slot production
```

Domain 3 limits reference

Resource	Limit
ARM template size	4 MB
Parameters per template	256
Resources per template	800
VMSS instances (marketplace images)	1,000
Availability Set fault domains	3
Availability Set update domains	20
Container Apps max replicas	30
App Service deployment slots (Premium)	20

Domain 4: Implement and manage virtual networking (15-20%)

Virtual networks and subnets

Azure reserved IP addresses (per subnet)

Azure reserves **5 IP addresses** in each subnet:

- x.x.x.0 — Network address
- x.x.x.1 — Default gateway
- x.x.x.2, x.x.x.3 — Azure DNS mapping
- x.x.x.255 — Broadcast address

Smallest supported subnet: /29 ([Microsoft Learn](#)) (provides 3 usable IPs)

VNet commands

```
bash
```

```
# Create VNet with subnet
```

```
az network vnet create --name vnet-1 --resource-group test-rg --address-prefix 10.0.0.0/16 \  
  --subnet-name subnet-1 --subnet-prefixes 10.0.0.0/24
```

```
# Add subnet to existing VNet
```

```
az network vnet subnet create --name subnet-2 --resource-group test-rg --vnet-name vnet-1 \  
  --address-prefixes 10.0.1.0/24
```

VNet peering

Key constraints:

- **Non-overlapping address spaces** required
- **Peering is NOT transitive** ($A \leftrightarrow B$ and $B \leftrightarrow C$ does NOT mean $A \leftrightarrow C$) ([Microsoftgeek](#))
- **Both peerings must be created** (one from each VNet)
- Basic Load Balancer cannot communicate across globally peered VNets

```
bash
```

```
# Create peering
```

```
az network vnet peering create --name vnet-1-to-vnet-2 --vnet-name vnet-1 \  
  --remote-vnet vnet-2 --resource-group test-rg --allow-vnet-access --allow-forwarded-traffic
```

Network Security Groups

NSG rule properties

Property	Details
Priority	100-4096 (lower number = higher priority)
Protocol	TCP, UDP, ICMP, ESP, AH, or Any
Direction	Inbound or Outbound
Action	Allow or Deny

Default rules (cannot be deleted)

Inbound:

Priority	Name	Source	Destination	Action
65000	AllowVNetInBound	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Deny

Outbound:

Priority	Name	Source	Destination	Action
65000	AllowVNetOutBound	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Deny

NSG processing order

Inbound: Subnet NSG → NIC NSG (traffic must be allowed by BOTH) [Microsoft Learn](#)

Outbound: NIC NSG → Subnet NSG [Microsoft Learn](#)

```
bash

# Create NSG
az network nsg create --name myNSG --resource-group myRG --location eastus

# Create NSG rule
az network nsg rule create --name AllowHTTP --nsg-name myNSG --resource-group myRG \
  --priority 100 --direction Inbound --access Allow --protocol Tcp \
  --source-address-prefixes '*' --destination-port-ranges 80 443
```

Common service tags

Tag	Description
VirtualNetwork	VNet + peered VNets + connected networks
AzureLoadBalancer	Azure infrastructure load balancer
Internet	Public internet addresses
Storage	Azure Storage service IPs
AzureActiveDirectory	Microsoft Entra ID IPs

Azure Bastion

Feature	Basic	Standard	Premium
Concurrent connections	25	50	50
Peered VNets	Yes	Yes	Yes
Custom ports	No	Yes	Yes
Host scaling	No	Yes (up to 50)	Yes
Native client	No	Yes	Yes
Session recording	No	No	Yes

Requirements:

- Subnet must be named **AzureBastionSubnet**
- Minimum size: **/26 or larger**
- Public IP: Standard SKU, Static allocation

Service endpoints vs Private endpoints

Feature	Service Endpoints	Private Endpoints
Traffic Path	Azure backbone (leaves VNet) Microsoft Community Hub	Stays within VNet Microsoft Learn
IP Address	Public IP of service Microsoft Learn	Private IP from VNet Microsoft Learn
DNS	Uses public DNS	Requires private DNS zones
On-premises access	Requires public IP whitelist	Works via VPN/ExpressRoute
Granularity	Subnet level	Resource instance level
Cost	Free	Per hour + data processing

```
bash

# Enable service endpoint on subnet
az network vnet subnet update --name mySubnet --vnet-name myVNet --resource-group myRG \
  --service-endpoints Microsoft.Storage Microsoft.Sql
```

Azure Load Balancer

Standard vs Basic SKU (Basic retired September 2025)

Feature	Basic (Retired)	Standard
Backend pool size	Up to 300	Up to 1,000
Health probes	TCP, HTTP	TCP, HTTP, HTTPS
Availability Zones	Not supported	Zone-redundant
Security	Open by default GitHub	Secure by default (needs NSG) Microsoft Learn
SLA	None	99.99%

```
bash

# Create public load balancer
az network lb create --name myLoadBalancer --resource-group myRG --sku Standard \
  --frontend-ip-name myFrontEnd --backend-pool-name myBackEndPool --public-ip-address myPubli

# Create health probe
az network lb probe create --name myHealthProbe --lb-name myLoadBalancer --resource-group my
  --protocol tcp --port 80 --interval 15 --threshold 2

# Create load balancing rule
az network lb rule create --name myLBRule --lb-name myLoadBalancer --resource-group myRG \
  --protocol tcp --frontend-port 80 --backend-port 80 --frontend-ip-name myFrontEnd \
  --backend-pool-name myBackEndPool --probe-name myHealthProbe
```

Health probe source IP: 168.63.129.16 (must allow in NSG) [Microsoft Learn](#)

Azure DNS

```
bash

# Create public DNS zone
az network dns zone create --name contoso.com --resource-group myRG

# Create A record
az network dns record-set a add-record --zone-name contoso.com --resource-group myRG \
  --record-set-name www --ipv4-address 10.10.10.10

# Create private DNS zone with VNet link
az network private-dns zone create --name private.contoso.com --resource-group myRG
az network private-dns link vnet create --name myVNetLink --zone-name private.contoso.com \
  --resource-group myRG --virtual-network myVNet --registration-enabled true
```

Domain 4 limits reference

Resource	Limit
VNets per subscription per region	1,000
Subnets per VNet	3,000
VNet peerings per VNet	500 (1,000 with AVNM) GitHub
NSGs per subscription per region	5,000
NSG rules per NSG	1,000
Routes per route table	400

Domain 5: Monitor and maintain Azure resources (10-15%)

Azure Monitor metrics and logs

Data retention

Data Type	Default Retention	Maximum
Platform Metrics	93 days	93 days
Log Analytics logs	31 days	730 days (2 years)
Application Insights logs	90 days	730 days

Essential KQL queries

kql

```
// VM CPU Usage (last hour)
Perf
| where ObjectName == "Processor" and CounterName == "% Processor Time"
| where TimeGenerated > ago(1h)
| summarize avg(CounterValue) by Computer, bin(TimeGenerated, 5m)
```

```
// VMs not sending heartbeats
Heartbeat
| summarize LastHeartbeat = max(TimeGenerated) by Computer
| where LastHeartbeat < ago(5m)
```

```
// Failed login attempts
SecurityEvent
| where EventID == 4625
| summarize count() by Account, Computer
```

```
// Azure resource deletions
AzureActivity
| where OperationNameValue contains "delete"
| where TimeGenerated > ago(7d)
```

Key KQL operators

Operator	Purpose	Example
where	Filter rows	<code>where TimeGenerated > ago(1h)</code>
project	Select columns	<code>project Computer, CounterValue</code>
summarize	Aggregate	<code>summarize count() by Computer</code>
extend	Create columns	<code>extend Duration = EndTime - StartTime</code>
join	Combine tables	<code>Table1 join Table2 on CommonColumn</code>
render	Visualize	<code>render timechart</code>

Alerts and action groups

Alert types

Alert Type	Based On	Response Time
Metric alerts	Metrics	Near real-time
Log search alerts	Log queries	Depends on frequency
Activity log alerts	Activity log	Near real-time

```
bash
```

```
# Create action group
```

```
az monitor action-group create --name MyActionGroup --resource-group MyRG \  
  --short-name MAG --action email admin admin@example.com
```

```
# Create metric alert rule
```

```
az monitor metrics alert create --name "High CPU Alert" --resource-group MyRG \  
  --scopes <resource-id> --condition "avg Percentage CPU > 80" \  
  --action MyActionGroup --window-size 5m --evaluation-frequency 1m
```

Alert processing rules

Apply modifications to triggered alerts at scale (maintenance windows, suppress non-business hours alerts).

```
bash
```

```
az monitor alert-processing-rule create --name 'AddActionGroupToSubscription' \  
  --rule-type AddActionGroups --scopes "/subscriptions/SUB1" \  
  --action-groups "/subscriptions/SUB1/resourcegroups/RG1/providers/microsoft.insights/actiongroups/AG1"
```

Network Watcher tools

Tool	Purpose
IP Flow Verify	Check if packet allowed/denied by NSG
Next Hop	Show next routing hop
Connection Troubleshoot	Test VM to endpoint connectivity
Connection Monitor	Continuous connectivity monitoring
Packet Capture	Capture network traffic
NSG Diagnostics	Analyze NSG rules

```
bash
```

IP Flow Verify

```
az network watcher test-ip-flow -g MyResourceGroup --direction Outbound --protocol TCP \  
  --local 10.0.0.4:* --remote 10.1.0.4:80 --vm MyVm
```

Next Hop

```
az network watcher show-next-hop -g MyResourceGroup --vm MyVm \  
  --source-ip 10.0.0.4 --dest-ip 10.1.0.4
```

Connection Troubleshoot

```
az network watcher test-connectivity --resource-group MyRG --source-resource MyVM \  
  --dest-address www.microsoft.com --dest-port 443
```

Azure Backup

Recovery Services Vault vs Backup Vault

Feature	Recovery Services Vault	Backup Vault
Workloads	Azure VMs, SQL in VMs, Azure Files	Azure Blobs, PostgreSQL, Disks
Soft Delete	14 days default	14 days default
Cross Region Restore	Yes (with GRS)	Yes (with GRS)

Backup commands

```
bash
```

Create Recovery Services vault

```
az backup vault create --resource-group MyRG --name MyRecoveryServicesVault --location eastus
```

Enable backup for VM

```
az backup protection enable-for-vm --resource-group MyRG --vault-name MyVault \  
  --vm MyVM --policy-name DefaultPolicy
```

Trigger on-demand backup

```
az backup protection backup-now --resource-group MyRG --vault-name MyVault \  
  --container-name MyVM --item-name MyVM --backup-management-type AzureIaaSVM \  
  --retain-until 01-01-2025
```

List recovery points

```
az backup recoverypoint list --resource-group MyRG --vault-name MyVault \  
  --container-name MyVM --item-name MyVM --backup-management-type AzureIaaSVM
```

Azure Site Recovery

Failover types

Type	Purpose	Data Loss
Test Failover	DR drill	None
Planned Failover	Scheduled migration	None
Failover	Disaster response	Depends on recovery point

Recovery point options

- **Latest:** Lowest RPO, processes all data first
- **Latest processed:** Low RTO, uses last processed point
- **Latest app-consistent:** Application-consistent recovery

```
powershell
```

```
# Test Failover
```

```
Start-AzRecoveryServicesAsrTestFailoverJob -ReplicationProtectedItem $protectedItem `
-Direction PrimaryToRecovery -RecoveryPoint $recoveryPoint
```

```
# Actual Failover
```

```
Start-AzRecoveryServicesAsrUnplannedFailoverJob -ReplicationProtectedItem $protectedItem `
-Direction PrimaryToRecovery
```

```
# Commit Failover
```

```
Start-AzRecoveryServicesAsrCommitFailoverJob -ReplicationProtectedItem $protectedItem
```

Domain 5 limits reference

Resource	Limit
Recovery Services vaults per region per subscription	500
Azure VMs per vault	1,000
Backup policies per vault	200
Metrics retention	93 days
Log Analytics max retention	730 days

Quick reference command table

Task	Azure CLI	PowerShell
Create user	<code>az ad user create</code>	<code>New-EntraUser</code>
Assign role	<code>az role assignment create</code>	<code>New-AzRoleAssignment</code>
Create policy assignment	<code>az policy assignment create</code>	<code>New-AzPolicyAssignment</code>
Create storage account	<code>az storage account create</code>	<code>New-AzStorageAccount</code>
Create VM	<code>az vm create</code>	<code>New-AzVM</code>
Create VMSS	<code>az vmss create</code>	<code>New-AzVmss</code>
Deploy ARM template	<code>az deployment group create</code>	<code>New-AzResourceGroupDeployment</code>
Create VNet	<code>az network vnet create</code>	<code>New-AzVirtualNetwork</code>
Create NSG	<code>az network nsg create</code>	<code>New-AzNetworkSecurityGroup</code>
Create Load Balancer	<code>az network lb create</code>	<code>New-AzLoadBalancer</code>
Create alert	<code>az monitor metrics alert create</code>	<code>Add-AzMetricAlertRuleV2</code>
Create backup vault	<code>az backup vault create</code>	<code>New-AzRecoveryServicesVault</code>

Exam day tips

The AZ-104 exam tests practical knowledge—focus on understanding **when** to use each service and **how** to configure them correctly. Key areas frequently tested include NSG rule priority evaluation, storage redundancy selection, RBAC scope inheritance, and backup/recovery procedures.

Critical concepts to master:

- NSG priority: **Lower number = higher priority** (100 processes before 200)
- Standard Load Balancer is **secure by default**—requires NSG to allow traffic
- VNet peering is **NOT transitive**—requires hub-spoke with gateway transit
- Archive tier not supported with ZRS redundancy options
- Resource locks apply to **management plane only**, not data plane
- Tags do **NOT inherit** from resource groups to resources
- AzureBastionSubnet must be **/26 or larger**

Practice with the Azure portal, CLI, and PowerShell to build hands-on familiarity with the configuration workflows tested on the exam.